

Data Stream V: Count Sketch Algorithm

COMPSCI 753

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RECAP: CountMin Sketch

- Query: find our estimation of $\tilde{f}_x$ using CountMin sketch matrix counter
- Answer: $\tilde{f}_x = \min(M_{1,h_1(x)}, M_{2,h_2(x)}, M_{3,h_3(x)}, M_{4,h_4(x)}) = 15$
- Query: given frequency threshold 20, is x a potential heavy hitter?
- Answer: No

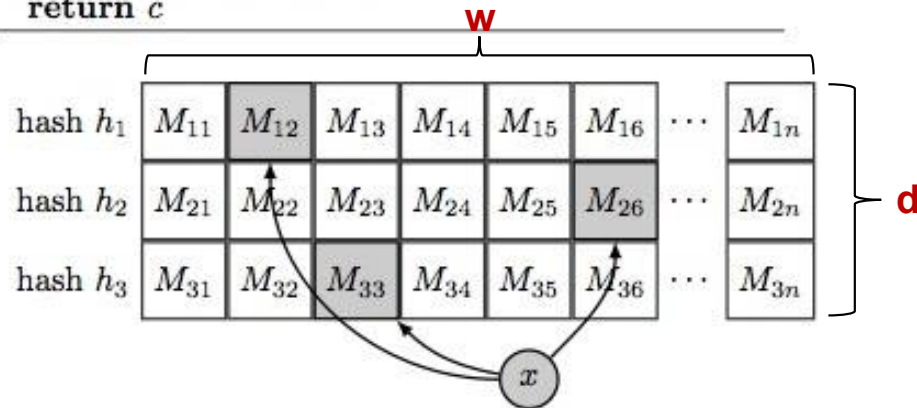
h_1	15	22	20	1	13	46	32	25	18	16	might collide with 1 item
h_2	10	16	14	13	30	19	45	23	7	31	might collide with 1 item
h_3	16	1	31	17	13	18	23	45	23	21	definitely collide with 2 items
h_4	17	22	3	31	6	15	54	6	25	29	might collide with 1 item

Randomized: CountMin Sketch

- Given
 - d independent **universal** hash functions h over range $[0, w)$
 - d different counters, $M_1, \dots, M_d$.
 - Each of size w initialized with 0s.
- Update to item x :
 - For each hash function, compute $h_i(x)$ and increment $M_{i,h_i(x)}$ by 1
- Query x
 - Return $\min(M_{1,h_1(x)}, \dots, M_{d,h_d(x)})$

Algorithm 1 Count-Min Sketch

```
insert( $x$ ):  
  for  $i = 1$  to  $d$  do  
     $M[i, h_i(x)] \leftarrow M[i, h_i(x)] + 1$   
  end for  
  
query( $x$ ):  
   $c = \min \{M[i, h_i(x)] \text{ for all } 1 \leq i \leq d\}$   
  return  $c$ 
```



Outline

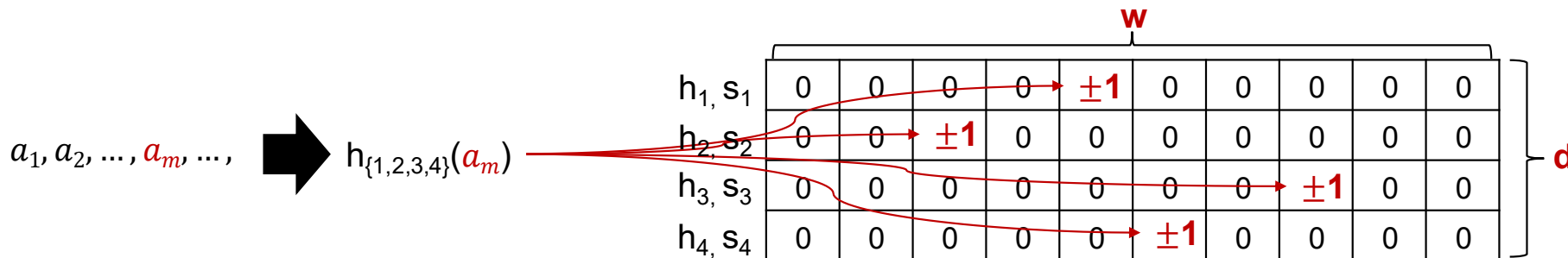
- Sampling from a Data Stream
- Filtering a Data Stream: Bloom Filters
- Finding Heavy Hitters in Data Stream
 - Misra-Gries Algorithm
 - CountMin Sketch Algorithm
 - Algorithm Illustration
 - Theoretical Bound
 - Count Sketch Algorithm
 - Algorithm Illustration
 - Theoretical Bound



Data as Infinite and Non-stationary

Counting Problem

- Count Sketch data structure: matrix of counters, each row corresponds to
 - 2-wise hash functions $\mathbf{h}$ over range $[0, w)$: Hash elements uniformly to $\{0, \dots, w - 1\}$
 - 2-wise hash functions $\mathbf{s}$ over range $\{+1, -1\}$: Assign the sign with probability 0.5
 - Initially set each counter to zero
- For each hash function, compute $\mathbf{h}_i(a)$ and increment $M_{i, \mathbf{h}_i(x)}$ by $\mathbf{s}_i(a)$



Example: Insertion, Update

$h_1(a_1)$	h_1, s_1	0	0	0	0	0 ⁺¹	0	0	0	0	0
$h_2(a_1)$	h_2, s_2	0	0	0	0 ⁻¹	0	0	0	0	0	0
$h_3(a_1)$	h_3, s_3	0	0	0	0	0	0	0 ⁺¹	0	0	0
$h_4(a_1)$	h_4, s_4	0	0	0	0	0	0	0	0	0	0 ⁻¹

$h_{\{1,2,3,4\}}(a_2)$	h_1, s_1	0	0	0 ⁻¹	0	1	0	0	0	0	0
	h_2, s_2	0	0 ⁺¹	0	-1	0	0	0	0	0	0
	h_3, s_3	0	0	0	0	0	0	0	0 ⁺¹	0	0
	h_4, s_4	0	0	0	0	0 ⁻¹	0	0	0	0	-1

Collision: $h_3(a_1) = h_3(a_2)$

Example: Query Heavy Hitters

- **Query:** find our estimation of $\tilde{f}_x$ using CountMin sketch matrix counter
- **Answer:** $\tilde{f}_x = \text{median}(s_1(x)M_{1,h_1(x)}, s_2(x)M_{2,h_2(x)}, s_3(x)M_{3,h_3(x)}, s_4(x)M_{4,h_4(x)}) = (10 + 12) = 11$
- **Query:** given frequency threshold 20, is x a potential heavy hitter?
- **Answer:** No

$h_{\{1,2,3,4\}}(x)$	h_1, s_1	15	2	10	1	13	23	16	25	1	16
	h_2, s_2	2	16	14	13	15	19	3	23	7	16
	h_3, s_3	16	1	11	17	13	8	12	22	11	21
	h_4, s_4	0	12	3	3	6	7	54	6	12	5

Randomized: Count Sketch

- Given

- d independent 2-wise hash functions h over range $[0, w)$
- d independent 2-wise hash functions s over range $\{-1, +1\}$
- d different counters, $M_1, \dots, M_d$
- Each of size w initialized with 0s

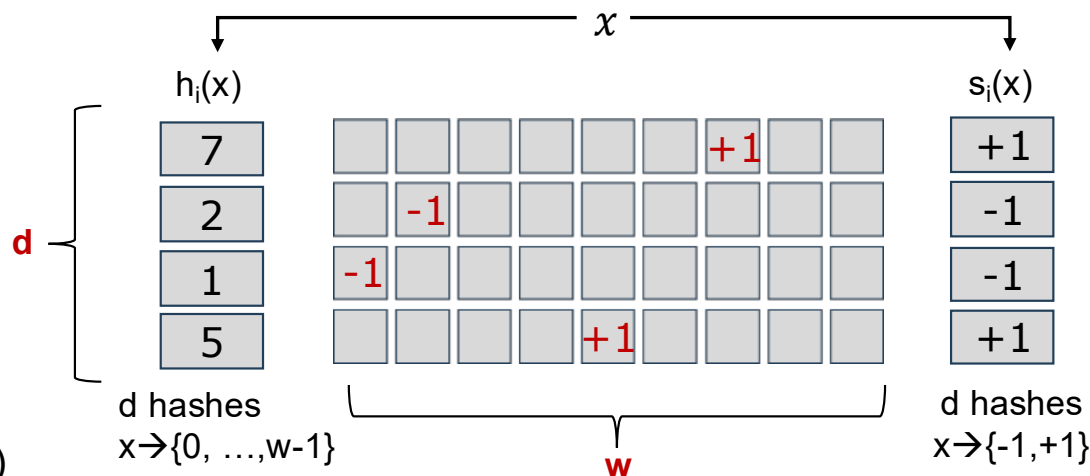
- Update to item x :

- For each hash function, compute $h_i(x)$ and increment $M_{i,h_i(x)}$ by $s_i(x)$

- Query x :

- Return **median**

$$(s_1(x)M_{1,h_1(x)}, \dots, s_d(x)M_{d,h_d(x)})$$



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Data as Infinite and Non-stationary

2-wise Hash Family

- 2-wise hash function definition
 - A family of hash function $H = \{h : U \rightarrow \{0, 1, \dots, w - 1\}\}$ is **2-wise independent** if for any 2 distinct keys $x_i \neq x_j \in U$ and **2 hash values** $y_i, y_j \in \{0, 1, \dots, w - 1\}$, the following holds:

$$\Pr_h[h(x_i) = y_i \wedge h(x_j) = y_j] = 1/w^2$$

- Count Sketch:** Given two different items $x_i \neq x_j$, what is the prob. x_i and x_j collide?

$$\Pr_h[h(x_i) = h(x_j)] = 1/w^2 \times w = 1/w$$

$\Pr_h[h(x_i) = h(x_j) = 0] = 1/w^2$	}	w
$\Pr_h[h(x_i) = h(x_j) = 1] = 1/w^2$		
...		
$\Pr_h[h(x_i) = h(x_j) = (w - 1)] = 1/w^2$		

Theoretical Claim

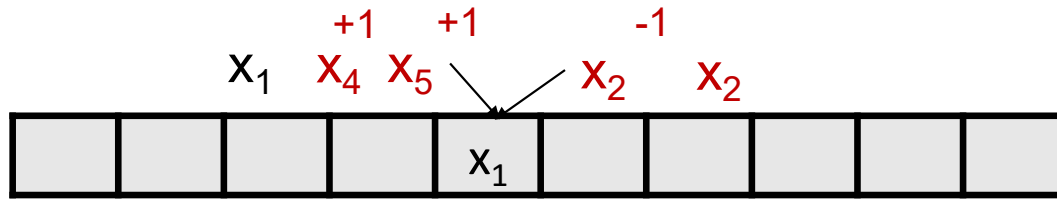
- **Parameter Setting:** $w = \frac{4}{\epsilon^2}$ and $d = 8\ln\left(\frac{1}{\delta}\right)$, where
 - ϵ is the error tolerance
 - m is the number of items in the stream
 - δ is the probability to fail
- **Property:** probability that estimated frequency is within a tolerable error approaches to 1

w									
15	22	20	1	13	46	32	25	18	16
10	16	14	13	30	19	45	23	7	31
16	1	31	17	13	18	23	45	23	21
17	22	3	31	6	15	54	6	25	29

d

Algorithmic Analysis: 1 Hash Function

- **1 hash function h .** Given a particular item x_1 , how many times $x_i \neq x_1$ collide by h ?



x_4 contributes miscounts by 1
 x_5 contributes miscounts by 1
 x_2 contributes miscounts by -2

- **Proof :**
 - Let $c_2, \dots, c_n$ be contributions of $x_2, \dots, x_n$ in the bucket $h(x_1)$
 - $$c_i = \begin{cases} -f_i, & \text{prob. } \frac{1}{2w} \\ 0, & \text{prob. } 1 - \frac{1}{w} \\ f_i, & \text{prob. } \frac{1}{2w} \end{cases}$$
 - $\leftarrow f_i$ collisions with each one collision prob. $1/w$ by 2-wise hash, $1/2$ "+" sign
 - $\leftarrow$ No collision
 - $\leftarrow f_i$ collisions with each one collision prob. $1/w$ by 2-wise hash, $1/2$ "-" sign
 - **Miscounts caused by the item x_i : $E[c_i] = 0$** $\leftarrow$ Expected miscount is zero. Our estimate is unbiased.

Algorithmic Analysis: 1 Hash Function

- Count Sketch Observations: Our estimation is **unbiased**
- Question: How many miscounts do we have? Can we apply Markov inequality?
- **No!** Markov inequality only applies to **non-negative** random variables.

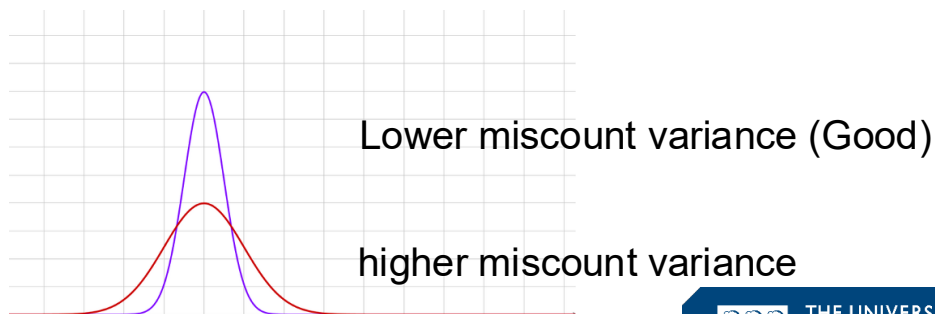
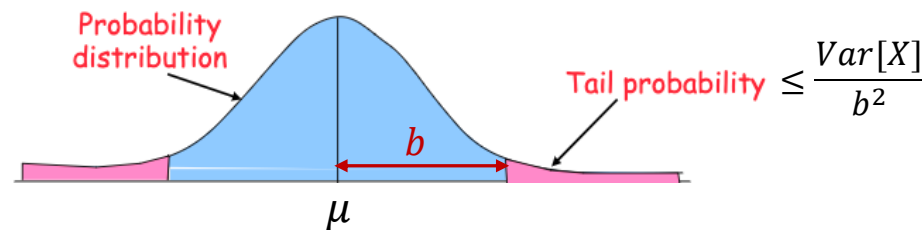
Markov: $\Pr(Z \geq a) \leq \frac{E[Z]}{a}$

Let $Z = (X - E[X])^2$
and $a = b^2$

$$\Pr((X - E[X])^2 \geq b^2) \leq \frac{\text{Var}[X]}{b^2}$$

μ : mean

Chebyshev: $\Pr(|X - \mu| \geq b) \leq \frac{\text{Var}[X]}{b^2}$



Algorithmic Analysis: 1 Hash Function

- Chebyshev's inequality: $\Pr(|X - \mu| \geq b) \leq \frac{\text{Var}[X]}{b^2}$
 - Let c_i be the contribution of x_i in the bucket $h(x_1)$

$$c_i = \begin{cases} -f_i, & \text{prob. } \frac{1}{2w} \\ 0, & \text{prob. } 1 - \frac{1}{w} \\ f_i, & \text{prob. } \frac{1}{2w} \end{cases} \longrightarrow c_i^2 = \begin{cases} f_i^2, & \text{prob. } \frac{1}{2w} \\ 0, & \text{prob. } 1 - \frac{1}{w} \\ f_i^2, & \text{prob. } \frac{1}{2w} \end{cases}$$

- Variance of miscounts: $\text{Var}[c_i] = E[c_i^2] - (E[c_i])^2 = \frac{f_i^2}{w}$
- Variance of total miscounts:

$$\text{Var}[Z] = \text{Var}[c_2] + \dots + \text{Var}[c_n] \leq \frac{f_2^2 + \dots + f_n^2}{w} \leq \frac{\sum_{i=1}^n f_i^2}{w} = \frac{\|f\|_2^2}{w} \quad \leftarrow f = (f_1, f_2, \dots, f_d)$$

- Using Chebyshev's inequality for $E[Z] = \mu$ and $b = \epsilon \|f\|_2$

$$\Pr(|Z| \geq \epsilon \|f\|_2) \leq \frac{\text{Var}[Z]}{\|f\|_2^2 \epsilon^2} \leq \frac{\|f\|_2^2}{w \|f\|_2^2 \epsilon^2} = \frac{1}{w \epsilon^2} = \frac{1}{4} \quad \leftarrow \text{if we choose } w = \frac{4}{\epsilon^2}$$

Miscount is within $\epsilon \|f\|_2$ with probability 1/4 (Good)

Algorithmic Analysis: d Hash Function

- **d hash functions $h_1, \dots, h_d$.** Given a particular item x_1 , the estimated frequency is as follows:

$$F_1 = \text{median}(s_1(x)M_{1,h_1(x_1)}, \dots, s_d(x)M_{d,h_d(x_1)}) = \text{median}(\tilde{f}_1, \tilde{f}_2, \dots, \tilde{f}_d)$$

- **Boosting the accuracy:** Using **d** independent hash functions corresponding to **d** independent arrays of counters.
 - Each estimator $\tilde{f}_i$ is good with more than 1/4 probability
 - The algorithm gets a bad estimation when more than half (i.e., $d/2$) of the estimators are bad (not within the range $\epsilon \|f\|_2$).

Algorithmic Analysis: d Hash Function

- **Chernoff bound:**

- If $X \sim \text{Binomial}(n, p)$ and $p < 1/2$, then

$$\Pr(X \geq n/2) \leq e^{-n(1/2-p)^2/2p}$$

- In our case, let Y be the number of **bad estimators**

- Then, $Y \sim \text{Binomial}(d, 1/4)$

$$\Pr(Y \geq d/2) \leq e^{-d/8}$$

- Let $d = 8 \ln \delta^{-1}$, then

$$\Pr(Y \geq d/2) \leq e^{-\frac{d}{8}} = \delta$$

- With probability at least **$1 - \delta$** , we have a probability that the miscount is within **$\epsilon \|f\|_2$**

Count Sketch Data Structure Size

$w = 4/\epsilon^2$

15	22	20	1	13	46	32	25	18	16
10	16	14	13	30	19	45	23	7	31
16	1	31	17	13	18	23	45	23	21
17	22	3	31	6	15	54	6	25	29

$d = 8\ln\left(\frac{1}{\delta}\right)$

- Count Sketch must contain $O(wd) = O(\epsilon^{-2} \log \delta^{-1})$ counters
- Example
 - $\epsilon = 10\%$ number of input data streams being miscounted, $\delta = 0.1 \rightarrow wd \approx 7300$ entries
 - $\epsilon = 1\%$ number of input data streams being miscounted, $\delta = 0.01 \rightarrow wd \approx 1.4\text{M}$ entries

Summary

- Randomization Technique for Finding Heavy Hitters
 - Count sketch produced closer estimated frequency compared to CountMin Sketch algorithm
 - Miscounts due to collisions
- Theoretic Bound
 - Good successful probably bound on tolerable miscount

Resources

- **[Book]** Mining of Massive Datasets by Jure Leskovec, Anand Rajaraman, and Jeff Ullman. Chapter 4
 - <http://mmds.org>
- **[Book Optional]** Machine Learning for Data Streams with Practical Examples in MOA (<https://moa.cms.waikato.ac.nz/book-html/>) – Chapter 4.6.3
- Multi-Purpose Sketches
 - <https://people.cs.umass.edu/~mcgregor/711S18/vectors-3.pdf>